

Better Gate — On-Board Inference

STM32F746G-DISCO TinyEngine INT8 · 96×96 leaf / not_leaf · full readable held-out test split · file tensors · no camera

95.14% STM32 accuracy 3,286 / 3,454	91.08% leaf recall 1,572 / 1,726	99.19% not_leaf recall 1,714 / 1,728	95.66% host INT8 (TFLite) 3,304 / 3,454	271 ms mean inference 269–273 ms
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What this run measures

This is a **file-on-MCU** accuracy check on the full readable test split. JPEG test images were resized to 96×96, ImageNet-normalized, and quantized with the Better Gate TFLite input scale (scale ≈ 0.01846, zero-point −16) into LE#####.RAW and NL#####.RAW (export_better_gate_stm32_raw.py --limit 0, seed 42). Firmware copied those bytes into leaf_gate_getInput() with no extra requantize and wrote RESULTS.CSV / SUMMARY.CSV to the SD card. There is no LCD, OV2640, or 176→96 camera path in this protocol.

Two leaf JPEGs in leaf_nonleaf_dataset/test are unreadable (fungal__leaf_spot186.jpg, viral__streak_virus118.jpg: broken data stream). They were skipped at export, so the board saw **1,726 leaf + 1,728 not_leaf = 3,454** files, not the 3,456 file listings in the training report. A prior 400-image subsample (200/class) is superseded by this run.

Headline result

On-board TinyEngine INT8 scored **3,286 / 3,454 = 95.14%**. training_report.pdf does not report INT8: its 95.78% (3,310 / 3,456) is **PyTorch FP32** on the full file listing. Host INT8 TFLite, using that report’s clean test transform (torchvision Resize 96×96 + ImageNet normalize) on the same 3,454 readable JPEGs, is **3,304 / 3,454 = 95.66%**. On-board is 0.52 pp below that host INT8 figure, not equal to it.

TinyEngine still matches TFLite when both see the **same RAW bytes**: re-running the INT8 TFLite on the host on the exported .RAW tensors produced identical labels on every file (3,454 / 3,454). The STM32 vs host-INT8 gap is the dump path (PIL resize + clip-to-int8) versus the training eval transform, not a broken MCU kernel.

Accuracy vs host training report

Evaluation	N	Overall	leaf recall	not_leaf recall	FN leaf	FP leaf
Host FP32 clean test (training_report.pdf)	3,456	95.78% 3,310 / 3,456	92.94% 1,606 / 1,728	98.61% 1,704 / 1,728	122	24
Host INT8 TFLite (same transform as that report)	3,454	95.66% 3,304 / 3,454	92.53% 1,597 / 1,726	98.78% 1,707 / 1,728	129	21
STM32 TinyEngine INT8 (RESULTS.CSV, PIL RAW dump)	3,454	95.14% 3,286 / 3,454	91.08% 1,572 / 1,726	99.19% 1,714 / 1,728	154	14

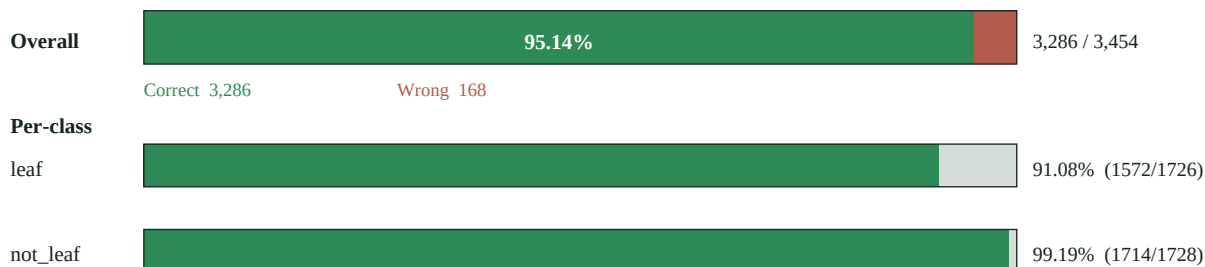
FP32 → host INT8 is a small drop (95.78% → 95.66% on readable files). Host INT8 → STM32 is another 0.52 pp (95.66% → 95.14%). The board is more conservative on leaf (FN 129 → 154) and slightly sharper on not_leaf (98.78% → 99.19%). Training eval used SafeImageFolder, which substitutes a neighbour JPEG for the two corrupt files so N stays 3,456; those two were never written as RAWs.

Accuracy graphs

Overall and per-class recall versus `training_report.pdf` FP32 and a host INT8 TFLite re-eval with that report's test transform. STM32 bars are the on-board RAW dump, not a copy of host INT8.



STM32 on-board (same as LCD graphs)



Confusion matrix — STM32 (rows = true)

true \ pred	leaf	not_leaf	n	recall
leaf	1,572	154	1,726	91.08%
not_leaf	14	1,714	1,728	99.19%
n pred / overall	1,586	1,868	3,454	95.14%

Precision, recall, F1

Class	Support	TP	FP	FN	Precision	Recall	F1
leaf	1,726	1,572	14	154	0.991	0.911	0.949
not_leaf	1,728	1,714	154	14	0.918	0.992	0.953

Firmware SUMMARY . CSV stores per-class accuracy as integers (leaf 91%, not_leaf 99%, overall 95%). This report uses the exact fractions from RESULTS . CSV.

Inference time

HAL-tick inference time is tight around **271 ms** (min 269, max 273, median 271, mean 270.8). That is the TinyLeafGate 96×96 INT8 cost on STM32F746, well below the 5-class disease CNN (~800–930 ms on the same MCU). The gate is cheap enough to sit in front of Nik's classifier without dominating the frame budget. Wall-clock for this 3,454-file run is dominated by the 1.5 s LCD result page, not by inference.

The 168 errors

All 168 MCU errors also occur when the host TFLite sees the same .RAW bytes. **154** are true leaf scored as not_leaf (gate would SKIP a real leaf). **14** are true not_leaf scored as leaf (disease CNN would run on junk). Mean `score_leaf` on those 154 misses is -36, versus +78 on correctly classified leaves. Of the leaf misses, 32 have `|score_leaf| ≤ 10`, 71 are in 11–40, 35

in 41–80, and 16 are confident ($|\text{score_leaf}| > 80$). Host INT8 with the training-report transform has fewer leaf misses (129 vs 154) because that path is not the PIL RAW dump.

The 14 false leaf calls ($\text{not_leaf} \rightarrow \text{leaf}$) are listed below. Several are near ties (NL0053 and NL0318 are 2/2 and 1/1). The rest are listed in `RESULTS.CSV` where `correct=0`.

File	score_leaf	score_not_leaf	File	score_leaf	score_not_leaf
NL0053.RAW	2	2	NL0820.RAW	24	−24
NL0149.RAW	4	0	NL0906.RAW	4	3
NL0171.RAW	36	−33	NL1068.RAW	57	−53
NL0209.RAW	23	−8	NL1111.RAW	6	3
NL0318.RAW	1	1	NL1169.RAW	29	−27
NL0400.RAW	57	−58	NL1239.RAW	47	−43
NL0577.RAW	61	−62	NL1253.RAW	15	−10

Scores are the firmware’s INT8 logits (not calibrated probabilities). Argmax is leaf vs not_leaf; ties resolve as leaf.

Setup

Item	Value
Model	Better Gate alt · TinyLeafGate 27,538 params · 96×96 · 2 class
Weights / TFLite	leaf_gate_output_alt / aclis_leaf_gate_96x_alt_full_int8.tflite
MCU / runtime	STM32F746G-DISCO · TinyEngine codegen (leaf_gate) · INT8
Input bytes	96×96×3 int8 = 27,648 · SD folder 0:/RAW/
Source images	leaf_nonleaf_dataset/test · 1,726 readable leaf + 1,728 not_leaf · seed 42
Skipped	2 corrupt leaf JPEGs (broken data stream); not written as .RAW
Host INT8	TFLite on JPEGs with training val_transform (torchvision Resize)
Export	export_better_gate_stm32_raw.py --limit 0 (PIL resize + int8 dump)
Logs	On-Board Inference/RESULTS.CSV · SUMMARY.CSV
Do not mix	5-class 176×176 BA/FU/... RAWs (different size and input scale)

What this does — and does not — claim

This run confirms TinyEngine on the Disco board reproduces host TFLite on the exported RAW tensors. It does **not** claim on-board file accuracy equals the training-report FP32 number, or the host INT8 number under that report’s torchvision test transform (95.66%). It does **not** claim the LCD+OV2640 AATBS numbers will rise. Run5 already showed that camera stills of the same PlantVillage JPEGs are a different distribution (flatten, undersample, panel glare, 176→96). Energy (PPK2) is still a separate measurement.

Source logs: Ikmal/Gate Model/Better Gate Model/On-Board Inference/RESULTS.CSV and SUMMARY.CSV.